Section 7.4 Exercises

Directions: For exercises 1 - 38, find all exact solutions on the interval $0 \leq \theta < 2\pi$.

1. $2\sin\theta = -\sqrt{2}$

2. $2\sin\theta = \sqrt{3}$

3. $2\cos\theta = 1$

4. $2\cos\theta = -\sqrt{2}$

5. $\tan\theta = -1$

6. $\tan x = 1$

7. $\cot x + 1 = 0$

8. $4\sin^2 x - 2 = 0$

9. $\csc^2 x - 4 = 0$

10. $2\cos\theta = \sqrt{2}$

11. $2\cos\theta = -1$

12. $2\sin\theta = -1$

13. $2\sin\theta = -\sqrt{3}$

14. $2\sin(3\theta) = 1$

15. $2\sin(2\theta) = \sqrt{3}$

16. $2\cos(3\theta) = -\sqrt{2}$

17. $\cos(2\theta) = -\frac{\sqrt{3}}{2}$

18. $2\sin(\pi\theta) = 1$

19. $2\cos\left(\frac{\pi}{5}\theta\right) = \sqrt{3}$

20. $\sec(x)\sin(x) - 2\sin(x) = 0$

21. $\tan(x) - 2\sin(x)\tan(x) = 0$

22. $2\cos^2 t + \cos(t) = 1$

23. $2\tan^2(t) = 3\sec(t)$

24. $2\sin(x)\cos(x) - \sin(x) + 2\cos(x) - 1 = 0$

25. $\cos^2\theta = \frac{1}{2}$

26. $\sec^2 x = 1$

27. $\tan^2(x) = -1 + 2\tan(-x)$

28. $\tan^5(x) = \tan(x)$

29. $\sin(2t) = \cos t$

30. $\cos(2t) = \sin t$

31. $\tan^2 x - \sqrt{3} \tan x = 0$

32. $\sin^2 x + \sin x - 2 = 0$

33. $\sin^2 x - \cos^2 x - \sin x = 0$

34. $\sin^2 x + \cos^2 x = 0$

35. $\sin(2x) - \sin x = 0$

36. $\cos(2x) - \cos x = 0$

37. $1 - \cos(2x) = 1 + \cos(2x)$

38. $\cos^3 t = \cos t$